An IoT-Enabled Hybrid Soft Sensor for Real-Time Residual-Moisture Monitoring and Resource Accounting in Compact Wool Scouring Machines

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***Abstract*—Wool scouring is among the most water- and energy-intensive steps in primary wool processing, yet small regional plants operate their machines without any instrumentation, and the key quality variable—residual moisture of the wool after each squeeze stage—is measured only occasionally by oven drying. This paper presents a low-cost Internet of Things (IoT) framework for a compact five-bowl wool scouring machine with an improved ribbed-drum/mesh squeezing device. An ESP32-based edge node acquires bowl temperature, liquor level and turbidity, squeeze-drum angular speed, make-up water flow and electrical power, and streams them over Wi-Fi (HTTP/MQTT) to a server that provides storage, alarms, per-kilogram resource indicators and a bilingual web/mobile application. We propose a hybrid soft sensor in which a design-of-experiments (DoE) regression model of the squeezing device, obtained from a 2^3 full-factorial experiment, serves as a physics-informed prior, and an online ridge-regression correction learned from sparse laboratory samples accounts for operating conditions. In a [TODO: N]-shift field deployment at [TODO: plant], the soft sensor achieved an RMSE of [TODO: x] % against oven-dry reference measurements, compared with [TODO: y] % for the DoE model alone and [TODO: z] % for a purely data-driven model, while the system delivered [TODO: packet-delivery ratio] % of messages with a median latency of [TODO: ms]. The measured specific energy and water consumption were [TODO: kWh/kg] and [TODO: L/kg].**

***Index Terms*—Industrial Internet of Things, soft sensor, wool scouring, edge computing, resource efficiency, textile manufacturing, digital twin.**

I. Introduction

PRIMARY wool processing—opening, scouring, squeezing and drying—determines the quality and cost of all downstream textile products. Scouring removes grease, suint and mineral dirt in a sequence of bowls filled with hot alkaline liquor; between bowls, squeezing devices remove the dirty liquor from the wool. Insufficient squeezing carries dirt into the next bowl, shortens liquor life and increases water and detergent consumption, and every percentage point of residual moisture must later be evaporated in the dryer. Scouring is a major contributor to the water footprint and life-cycle burden of wool products [3], [4], and its highly polluted effluent requires costly treatment [5], [6].

In regions such as Uzbekistan, where wool comes largely from Karakul sheep and is processed in small, decentralised plants, large centralised scouring lines are uneconomic. A compact, resource-saving machine with an improved squeezing device (ribbed squeeze drum, guide slat drum and wrapped mesh surface) was developed and patented in previous work [1], [2]; factory tests reported a reduction of residual moisture from 65–70 % to 55–60 % and of installed power from 13 to 4.2 kW. However, such machines are operated without sensors: operators cannot see liquor temperature or dirtiness, moisture is measured only by occasional oven-dry samples, and resource consumption is not accounted per kilogram of wool.

Related work. Industrial IoT increasingly moves analytics to the edge of the network [7], [8], and IoT/big-data frameworks have been demonstrated for health-state monitoring, fault detection and cloud–edge–device collaborative monitoring of manufacturing systems [9]–[11]. Low-cost nodes make such monitoring affordable [12], [13], and digital retrofitting is the practical route for legacy machines that cannot be replaced [14]. Digital twins built from sensor data extend this to model-based operation [15]–[17]. In the textile sector, Industry 4.0 and AI work concentrates on spinning, inspection and garment production [18]–[21]; energy benchmarking of wet processing in SMEs has been proposed but relies on manual audits [22].

Soft sensors estimate hard-to-measure quality variables from routine measurements [23]. Purely data-driven models need many labelled samples, which are expensive when the reference is a laboratory analysis [24]; gray-box and physics-informed models reduce this need and improve explainability [25], [26], while adaptive schemes keep the model calibrated online [27]. In IoT settings, self-improving soft sensors have been used to reduce costly measurements [28]. To our knowledge, however, no work instruments the wool scouring/squeezing stage, and none uses a design-of-experiments model of the machine as the prior of an IoT soft sensor.

Contributions.

1) A low-cost (≈ US$ 200) IoT architecture for wool scouring machines with an ESP32 edge node, HTTP/MQTT transport, server-side analytics and an installable bilingual (EN/RU) web/mobile application.
2) A hybrid soft sensor for residual moisture that combines a 2^3 factorial regression model of the squeezing device (geometry prior) with an online, regularised correction

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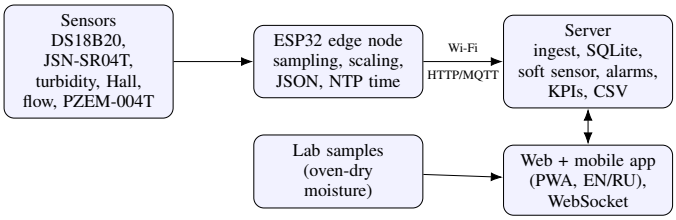


Fig. 1. Architecture of the proposed IoT framework.

for operating conditions, calibrated from sparse laboratory samples.

3) Real-time per-kilogram resource indicators (kWh/kg, L/kg) and rule-based fault alarms (heater failure, low level, dirty liquor, drum speed deviation).
4) A field deployment on an industrial machine with [TODO: N] shifts of data, with open code at https://github.com/nasin0764-bit/woolwash-iot and [TODO: open dataset].

II. Machine and Squeezing Model

The machine comprises five bowls (2.5 m^3 each); moving rakes (length 250 mm, linear speed [TODO: 0.15] m/s) transport the wool, and each bowl is followed by a squeezing device with a ribbed drum of diameter D, a mesh of hole diameter d and a drum–mesh gap h [1].

A. DoE prior

A 2^3 full-factorial experiment with three replicates ($N = 8$, $m = 3$) with coded factors $X_1 = (d - 2)/1$, $X_2 = (D - 250)/50$, $X_3 = (h - 3)/1$ (mm) gave, after Student's test ($t_{0.05;16} = 2.12$), the adequate model ($F = 0.71 < F_{0.05;4;16} = 3.01$)

$$W_0 = 57.31 + 0.81X_1 - 0.95X_2 + 0.30X_3 \quad [\%]. \tag{1}$$

B. Hybrid soft sensor

Equation (1) captures geometry only. During operation, moisture also depends on the drum angular speed ω, the feed rate q, the liquor temperature T and the liquor turbidity τ. With normalised deviations $\boldsymbol{\phi} = [(\omega - \omega_0)/s_\omega, (q - q_0)/s_q, (T - T_0)/s_T, (\tau - \tau_0)/s_\tau]^\top$ we model

$$\hat{W} = W_0(d, D, h) + \mathbf{w}^\top \boldsymbol{\phi} + c, \tag{2}$$

where $(\mathbf{w}, c)$ minimise $\sum_i (W_i - W_{0,i} - \mathbf{w}^\top \boldsymbol{\phi}_i - c)^2 + \lambda \|\mathbf{w}\|^2$ over laboratory samples W_i taken after the squeezer and dried at 105 °C. The model is refitted on the server each time a sample is added; with no samples it reduces to the DoE prior. [TODO: Compare with: DoE only; ridge/GPR/random forest without prior; report learning curve vs number of samples.]

III. IoT System Architecture

Fig. 1 shows the architecture. The node supports both HTTP and MQTT; the choice trades broker infrastructure against energy and overhead, as quantified in recent protocol comparisons [29], [30]. [TODO: Describe sensor placement (photo), sampling period (2 s), payload size, power supply, enclosure, calibration of each sensor (method and uncertainty).]

TABLE I
Historical 2^3 factorial data for residual moisture W (%) [1]. Levels $-1/0/+1$: $d = 1/2/3$ mm, $D = 200/250/300$ mm, $h = 2/3/4$ mm

Run	X_1	X_2	X_3	Replicates	Mean ± SD	$\hat{W}$
1	−	−	−	57.0, 57.5, 58.3	57.60 ± 0.66	57.15
2	+	−	−	58.3, 58.5, 58.7	58.50 ± 0.20	58.78
3	−	+	−	54.5, 55.2, 55.4	55.03 ± 0.47	55.26
4	+	+	−	57.2, 56.2, 57.4	56.93 ± 0.64	56.88
5	−	−	+	58.5, 57.2, 56.8	57.50 ± 0.89	57.74
6	+	−	+	59.8, 59.5, 59.0	59.43 ± 0.40	59.37
7	−	+	+	55.1, 56.3, 56.2	55.87 ± 0.67	55.85
8	+	+	+	57.1, 58.8, 57.0	57.63 ± 1.01	57.48

IV. Experimental Setup

[TODO: Plant, dates, wool type (spring/autumn Karakul), number of shifts, lab sampling protocol (every 30 min, 3 replicates), train/test split by day, network environment (AP distance, RSSI).]

V. Results

A. Retrospective mechanical-process baseline

The values in this subsection are transcribed from the historical experimental program [1] (Tables 3.7, 3.9, 3.13 and Sec. 4.1 therein) and reanalysed; they are *not* measurements of the IoT system and serve only as the physical-process reference and as the prior of the soft sensor.

Factorial experiment. The 2^3 design (Table I) comprised three residual-moisture measurements per treatment, giving 24 observations over eight geometry combinations. Unweighted least squares on the eight run means gives the main-effects model (1) ($b_0 = 57.3125$, $b_1 = 0.8125$, $b_2 = -0.9458$, $b_3 = 0.2958$); the interaction terms were not significant ($|t| \le 1.08 < 2.12$). The within-run variances satisfy Cochran's homogeneity criterion ($G = 0.290 < 0.516$), and the model passes the lack-of-fit test ($F = 0.714 < F_{0.05;4,16} = 3.01$). These tests indicate no lack of fit at the eight measured corners but do not establish predictive accuracy outside the factor domain. The selected 2 mm/250 mm/3 mm geometry is an engineering compromise—narrower gaps caused wool jamming and damage, wider gaps reduced squeezing—and corresponds to a *predicted* residual moisture of 57.31 %; no separate center-point run was performed, and the minimum fitted corner (run 3, 55.26 %) is lower, so the setting is not the unconstrained statistical optimum.

Historical industrial observations. The improved design was tested at two enterprises in Uzbekistan (Qashqadaryo and Surxondaryo regions) [1]. Table II summarises the reported values and midpoint-based normalisations. Post-squeezing moisture decreased by about 10 percentage points and water use per shift by 76.5 %, but the improved line also had lower nominal throughput and lower clean-wool yield; the latter may reflect better impurity removal or fibre loss and cannot be resolved from aggregate values. Installed power is a nameplate value, not metered energy. The records lack matched batches, per-site variability and uncertainty, and are

TABLE II
HISTORICAL COMPARISON REPORTED IN [1] (NOT IOT DATA)

Indicator	Existing	Improved	Change
Residual moisture (%)	65–70	55–60	−10 pp[a]
Water (m^3/shift)[b]	59.6	14.0	−76.5 %
Water reused (%)	70	80	+10 pp
Throughput (kg/h)[c]	700–800	400–450	−43.3 %[a]
Clean-wool yield (%)	64–66	60–62	−4 pp[a]
Installed power (kW)	13.0	4.2	−67.7 %
Water/throughput[d]	–	–	−58.5 %
Power/throughput[d]	–	–	−43.0 %
Scouring eff. (%)[e]	n/r	94.5; 92–95	–

[a] Range midpoints. [b] Units as in Table 3.13 of [1]; another table of the source uses litres per 450 kg. [c] Printed as "kg/s" in parts of the source; kg/h is the internally consistent reading. [d] Equal shift length and midpoint throughput assumed; nameplate power, not metered specific consumption. [e] Spring; autumn wool. n/r: not reported.

TABLE III
RESIDUAL-MOISTURE PREDICTION ON HELD-OUT FIELD SAMPLES

Model	RMSE (%)	MAE (%)	R^2
DoE prior only (1)	[TODO:]	[TODO:]	[TODO:]
Data-driven ridge (no prior)	[TODO:]	[TODO:]	[TODO:]
Random forest (no prior)	[TODO:]	[TODO:]	[TODO:]
Hybrid (proposed)	[TODO:]	[TODO:]	[TODO:]

therefore descriptive evidence only—which is precisely the measurement gap the proposed IoT system addresses.

B. Soft-sensor accuracy

C. Resource indicators

[TODO: kWh/kg and L/kg per shift, baseline vs improved machine, mean ± SD, n.]

D. Communication performance

[TODO: Packet delivery ratio, latency distribution, outages, energy of node.]

E. Fault detection

[TODO: Detection delay and false-alarm rate for heater-off, low-level, drum-speed scenarios.]

VI. DISCUSSION AND LIMITATIONS

Validation boundary. The historical data of Sec. V-A provide a physical-process reference but do not evaluate the sensing, networking or analytics of the proposed system. Only the prospective measurements reported in the remaining subsections of Sec. V—sensor calibration errors, delivery ratio and latency, and baseline-versus-IoT operating outcomes under matched conditions—support claims about the IoT architecture. [TODO: Generalisation across wool types, sensor fouling (turbidity probes in alkaline liquor), Wi-Fi in wet halls, scaling to multiple plants, security (device token, TLS).]

VII. CONCLUSION

[TODO:]

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